

$\mathcal{K}(\ell_p(\Lambda))$ is relatively injective as a right $\mathcal{B}(\ell_p(\Lambda))$ -module

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Abstract: In this note we prove a somewhat unexpected result: the Banach space of compact operators on a reflexive ℓ_p -space is relatively injective as a right Banach module over the algebra of bounded linear operators on the same space.

Keywords: injective Banach module, compact operators, ℓ_p -space.

1 Introduction

Extension problems have been an important topic of functional analysis since its inception. Such a problem asks whether a morphism defined on a subobject can be extended to the whole object; the objects into which every morphism extends are precisely the injective ones. Injectivity is therefore the categorical counterpart of the solvability of extension problems. For instance, the field of complex numbers is an injective object in the category of Banach spaces. All known injective Banach spaces are isomorphic to the space of continuous functions on some compact space [1]. The property of the object in a given category to be injective depends on the “size” of the category and the “abundance” of morphisms. For example, if one restricts the category of Banach spaces to separable ones, the space c_0 of vanishing sequences becomes injective [2]. Moreover, Zippin showed in [3] that all injective spaces in the category of separable Banach spaces are isomorphic to c_0 . Similarly, if one restricts the set of morphisms to module maps, without changing the class of objects, then for any set Λ the space $c_0(\Lambda)$ is relatively injective as a right $\ell_\infty(\Lambda)$ -module [4]. These results rely on specific Banach-geometric properties of c_0 -spaces. One might therefore expect that the non-commutative counterpart of these results fails: that the space of compact operators on a Hilbert space $\mathcal{K}(H)$ is not injective, its geometry being quite different. Nevertheless the result holds, not only for Hilbert spaces, but for all reflexive ℓ_p -spaces.

2 Notation and definitions

Before we proceed to the main topic, we fix some notation and recall a few definitions.

If M is a subset of a set N , then χ_M denotes the indicator function of M . If $f : N \rightarrow L$ is an arbitrary function, then $f|_M$ denotes its restriction to M .

For given Banach spaces E and F , by $\mathcal{B}(E, F)$ we denote the space of all bounded linear operators from E to F , and by $\mathcal{K}(E, F)$ the space of compact operators from E to F . We use the abbreviations $\mathcal{B}(E) := \mathcal{B}(E, E)$ and $\mathcal{K}(E) := \mathcal{K}(E, E)$. The symbol 1_E stands for the identity operator on E . By $\langle \cdot, \cdot \rangle : E \times E^* \rightarrow \mathbb{C}$, $(x, f) \mapsto f(x)$ we denote the standard duality between E and E^* . For given $x \in F$ and $f \in E^*$, by $x \circ f$ we denote the rank-one operator from E to F defined by the equality $(x \circ f)(y) = \langle y, f \rangle x$. One can easily check that $\|x \circ f\| = \|x\| \|f\|$.

For an infinite set Λ we write $c_0(\Lambda)$ for the space of all functions $x : \Lambda \rightarrow \mathbb{C}$ such that for every $\varepsilon > 0$ the set $\{\lambda \in \Lambda : |x(\lambda)| \geq \varepsilon\}$ is finite, and $\ell_\infty(\Lambda)$ for the space of all bounded such functions. For $p \in [1, +\infty)$ the space $\ell_p(\Lambda)$ consists of all functions $x : \Lambda \rightarrow \mathbb{C}$ with finite ℓ_p -norm $\|x\|_p := (\sum_{\lambda \in \Lambda} |x(\lambda)|^p)^{1/p}$. For a family of Banach spaces $\{E_\lambda : \lambda \in \Lambda\}$ and $p \in [1, +\infty]$ by $\bigoplus_p \{E_\lambda : \lambda \in \Lambda\}$ we denote their ℓ_p -sum. Clearly, $\ell_p(\Lambda) = \bigoplus_p \{\mathbb{C} : \lambda \in \Lambda\}$.

A bounded linear functional $\mu \in \ell_\infty(\Lambda)^*$ is called a *singular state* on $\ell_\infty(\Lambda)$ if (i) $\|\mu\| = 1$; (ii) $\mu(\chi_\Lambda) = 1$; and (iii) $\mu|_{c_0(\Lambda)} = 0$. Conditions (i) and (ii) say that μ is a state on the unital Banach algebra $\ell_\infty(\Lambda)$, while (iii) says that μ vanishes on $c_0(\Lambda)$. Singular states exist for every infinite Λ : the quotient $\ell_\infty(\Lambda)/c_0(\Lambda)$ is a unital commutative C^* -algebra, hence carries a state σ (for instance the evaluation at any point of its maximal ideal space), and the composition of σ with the quotient homomorphism $\ell_\infty(\Lambda) \rightarrow \ell_\infty(\Lambda)/c_0(\Lambda)$ is a singular state on $\ell_\infty(\Lambda)$. We do not require μ to be multiplicative.

Now we shall recall a few notions from the geometry of Banach spaces. A sequence $(x_n)_{n \in \mathbb{N}}$ in a Banach space E is called *weakly null*, *weak^{*} null* or *norm null* if it converges to zero weakly, weakly^{*} or in norm, respectively. The sequence $(x_n)_{n \in \mathbb{N}}$ is called *semi-normalized* if the norms of its elements have a positive lower bound and a finite upper bound. Let $(\Lambda, <)$ be a well-ordered set. A family of vectors $(e_\lambda)_{\lambda \in \Lambda}$ is called a *Schauder basis* of E if for any vector $x \in E$ there exists a unique family of scalars $(f_\lambda(x))_{\lambda \in \Lambda} \subset \mathbb{C}$ such that

$$x = \sum_{\lambda \in \Lambda} f_\lambda(x) e_\lambda.$$

In general the value of this sum may depend on the ordering of Λ ; if it does not, the basis is called *unconditional*. One can show [[7], theorem 1.1.3] that the maps $(f_\lambda)_{\lambda \in \Lambda}$ are well-defined bounded linear functionals. They are called the *biorthogonal functionals* of $(e_\lambda)_{\lambda \in \Lambda}$.

A sequence of vectors $(e_n)_{n \in \mathbb{N}}$ in E is called a *basic sequence* if it is a Schauder basis of its closed linear span. The following construction gives an important source of basic sequences. Suppose $(e_\lambda)_{\lambda \in \Lambda}$ is a Schauder basis of some Banach space E , and let $(\Lambda_n)_{n \in \mathbb{N}}$ be a sequence of disjoint finite subsets of Λ such that $\max(\Lambda_n) < \min(\Lambda_{n+1})$ for all $n \in \mathbb{N}$. A sequence of non-zero vectors $(u_n)_{n \in \mathbb{N}}$ such that u_n lies in the linear span of $(e_\lambda)_{\lambda \in \Lambda_n}$ is called a *block-basic sequence*. Such sequences are indeed basic [[7], lemma 1.3.5]. If E and F are two Banach spaces, then two sequences $(u_n)_{n \in \mathbb{N}} \subset E$ and $(v_n)_{n \in \mathbb{N}} \subset F$ are called *equivalent* if there is an isomorphism of Banach spaces $T : E \rightarrow F$ such that $T(u_n) = v_n$ for all $n \in \mathbb{N}$.

Finally, we give a short introduction to Banach homology theory. Let A be a Banach algebra. We shall consider only right Banach modules over A with contractive outer action $\cdot : X \times A \rightarrow X$. If X and Y are two right Banach A -modules, then a map $\phi : X \rightarrow Y$ is an *A -morphism* if it is a continuous A -module map. Banach A -modules and A -morphisms form a category which we denote by **mod** $- A$.

In **mod** $- A$ the notion of injectivity can be defined in different ways. We shall be concerned with relative injectivity. Let $\xi : X \rightarrow Y$ be an A -morphism. Then it is called *c -relatively admissible* if $T \circ \xi = 1_X$ for some bounded linear operator $T : Y \rightarrow X$ with $\|T\| \leq c$. A Banach A -module J is called *C -relatively injective* if for any c -relatively admissible A -morphism $\xi : X \rightarrow Y$ and any A -morphism $\phi : X \rightarrow J$ there exists a continuous A -module map $\psi : Y \rightarrow J$ such that $\|\psi\| \leq cC\|\phi\|$ and the diagram

$$\begin{array}{ccc} & & Y \\ & \nearrow \psi & \uparrow \xi \\ J & \xleftarrow{\phi} & X \end{array}$$

commutes. A right Banach A -module is called *relatively injective* if it is C -relatively injective for some $C \geq 1$. As an example, if $A = \{0\}$, the category **mod** $- A$ reduces to the ordinary category of Banach spaces, and in this case all Banach spaces are relatively injective.

All injectivity statements in this note are verified by means of a single criterion, due to Helemskii, which reduces relative injectivity to the existence of one module retraction. Suppose A is a

unital Banach algebra acting on itself by multiplication, and let J be a right Banach A -module. The space $\mathcal{B}(A, J)$ of bounded linear operators from A to J is a right A -module under the action $(\mathcal{T} \cdot a)(b) := \mathcal{T}(ab)$; it is the *cofree* module cogenerated by J . There is a canonical A -morphism

$$\rho_J : J \rightarrow \mathcal{B}(A, J), \quad x \mapsto (a \mapsto x \cdot a),$$

and ρ_J always admits a bounded linear (not necessarily module) left inverse, namely $\mathcal{T} \mapsto \mathcal{T}(1_A)$, so it is relatively admissible. Helemskii's criterion [[6], proposition VII.1.39(III)] states that J is relatively injective if and only if ρ_J admits a left inverse *in the category $\mathbf{mod} - A$* , that is, a right A -module map $\tau : \mathcal{B}(A, J) \rightarrow J$ with $\tau \circ \rho_J = 1_J$. Moreover, by a theorem of White [[8], proposition 3.8], if such a τ can be chosen contractive then J is 1-relatively injective.

3 Main result

In this section we shall prove that for any reflexive ℓ_p -space the right Banach $\mathcal{B}(\ell_p(\Lambda))$ -module $\mathcal{K}(\ell_p(\Lambda))$ is relatively injective.

Before turning to compact operators, we recall that the ambient algebra is always relatively injective as a module over itself, and explain why the passage to the ideal of compact operators is harder.

Proposition 3.1. *For any Banach space E the right $\mathcal{B}(E)$ -module $\mathcal{B}(E)$ is 1-relatively injective.*

Proof. If $E = \{0\}$ then $\mathcal{B}(E) = \{0\}$ and the claim is trivial, so assume $E \neq \{0\}$. Fix $e_0 \in E$ and $f_0 \in E^*$ with $\langle e_0, f_0 \rangle = 1$ and $\|e_0\| = \|f_0\| = 1$, and define

$$\tau : \mathcal{B}(\mathcal{B}(E), \mathcal{B}(E)) \rightarrow \mathcal{B}(E), \quad \mathcal{T} \mapsto (x \mapsto \mathcal{T}(x \circ f_0)(e_0)).$$

Using the identity $A \circ (x \circ f_0) = (Ax) \circ f_0$ together with $\langle e_0, f_0 \rangle = 1$, it is trivial to check that τ is a contractive right $\mathcal{B}(E)$ -module map which is a left inverse for the canonical morphism $\rho_{\mathcal{B}(E)}$. By Helemskii's criterion and White's theorem, recalled in Section 2, the module $\mathcal{B}(E)$ is 1-relatively injective. $\square$

The single evaluation works because $\tau(\mathcal{T})$ may be any bounded operator. This fails in the compact case. If $\mathcal{T} : \mathcal{B}(E) \rightarrow \mathcal{K}(E)$ and we keep the same formula, then $\tau(\mathcal{T})$ need not be compact: for $\mathcal{T}(S) := (Se_0) \circ f_0$ the formula gives $\tau(\mathcal{T}) = 1_E$, which is non-compact in infinite dimensions. To keep the value in $\mathcal{K}(E)$ we average the evaluation over a whole biorthogonal system instead of reading it off at one point. The singular state below provides this average.

There is also an a priori obstruction that explains why compactness cannot be taken for granted. We record it as a necessary condition for the ideal of compact operators.

Proposition 3.2. *Let E be an infinite-dimensional Banach space. If the right $\mathcal{B}(E)$ -module $\mathcal{K}(E)$ is relatively injective, then $\mathcal{K}(E)$ is not complemented in $\mathcal{B}(E)$ as a Banach space.*

Proof. The ideal $\mathcal{K}(E)$ has no left identity: if $u \in \mathcal{K}(E)$ satisfied $u \circ K = K$ for all $K \in \mathcal{K}(E)$, then testing on the rank-one operators $K = x \circ f$ would give $u(x) = x$ for all $x \in E$, that is $u = 1_E$, which is not compact in infinite dimensions. On the other hand, by a result of Ramsden [[9], proposition 2.2.8(iii)], a relatively injective closed right ideal of a Banach algebra which is complemented as a subspace possesses a left identity. Were $\mathcal{K}(E)$ relatively injective and complemented in $\mathcal{B}(E)$, this would furnish a left identity for $\mathcal{K}(E)$, a contradiction. $\square$

Proposition 3.2 is a necessary condition only, and a weak one, since $\mathcal{K}(E)$ is uncomplemented in $\mathcal{B}(E)$ for most classical E . It is not vacuous, however. If the Calkin algebra $\mathcal{B}(E)/\mathcal{K}(E)$ is finite-dimensional, then $\mathcal{K}(E)$ is finite-codimensional and hence complemented, so by Proposition 3.2 its module $\mathcal{K}(E)$ is not relatively injective. Such spaces exist: on the Argyros-Haydon space $\mathfrak{X}_{AH}$ every operator is a scalar plus a compact one, so the Calkin algebra is one-dimensional [10]; Tarbard constructed preduals of ℓ_1 whose Calkin algebra is finite-dimensional of any prescribed dimension [11]. The point of the proposition is that relative injectivity of $\mathcal{K}(E)$ cannot come from a bounded projection and must be witnessed by a module splitting of the cofree module. We construct such a splitting for reflexive ℓ_p -spaces in the rest of this section.

Proposition 3.3. *Let Λ be an infinite set, and μ be a singular state on $\ell_\infty(\Lambda)$. If $|\mu(x)| \geq c$ for some $x \in \ell_\infty(\Lambda)$ and $c > 0$, then for any $c' < c$ the set $\{\lambda \in \Lambda : |x(\lambda)| > c'\}$ is infinite.*

Proof. Suppose to the contrary that for some $c' < c$ the set $F := \{\lambda \in \Lambda : |x(\lambda)| > c'\}$ is finite. Decompose $x = x\chi_F + x\chi_{\Lambda \setminus F}$. Since F is finite, $x\chi_F \in c_0(\Lambda)$, hence $\mu(x\chi_F) = 0$ by property (iii) of a singular state. On the complement we have $\|x\chi_{\Lambda \setminus F}\| \leq c'$, so

$$|\mu(x)| = |\mu(x\chi_F) + \mu(x\chi_{\Lambda \setminus F})| = |\mu(x\chi_{\Lambda \setminus F})| \leq \|\mu\| \|x\chi_{\Lambda \setminus F}\| \leq c' < c,$$

which contradicts $|\mu(x)| \geq c$. Therefore F is infinite. $\square$

Proposition 3.4. *Let E be a reflexive Banach space, let Λ be an infinite set, and let μ be a singular state on $\ell_\infty(\Lambda)$. Then there exists a contractive linear operator $M_\mu : \bigoplus_\infty \{E : \lambda \in \Lambda\} \rightarrow E$ such that:*

- (i) $\langle M_\mu(\xi), f \rangle = \mu(\bigoplus_\infty \{\langle \xi(\lambda), f \rangle : \lambda \in \Lambda\})$ for all $\xi \in \bigoplus_\infty \{E : \lambda \in \Lambda\}$ and $f \in E^*$;
- (ii) $M_\mu(\bigoplus_\infty \{x : \lambda \in \Lambda\}) = x$ for any $x \in E$;

Proof. Let $\xi \in \bigoplus_\infty \{E : \lambda \in \Lambda\}$, then $\{\langle \xi(\lambda), f \rangle : \lambda \in \Lambda\} \subset \mathbb{C}$ is bounded, so the map

$$F_{\mu, \xi} : E^* \rightarrow \mathbb{C}, f \mapsto \mu\left(\bigoplus_\infty \{\langle \xi(\lambda), f \rangle : \lambda \in \Lambda\}\right)$$

is well defined. Clearly, it is a bounded linear functional with $\|F_{\mu, \xi}\| \leq \|\xi\|$. Since E is reflexive, there exists a unique vector $M_{\mu, \xi} \in E$ such that $\|M_{\mu, \xi}\| = \|F_{\mu, \xi}\|$ and $F_{\mu, \xi}(f) = \langle M_{\mu, \xi}, f \rangle$ for all $f \in E^*$. Consider the map $M_\mu : \bigoplus_\infty \{E : \lambda \in \Lambda\} \rightarrow E$, $\xi \mapsto M_{\mu, \xi}$. It is easy to check that M_μ is a linear operator. It is contractive because $\|M_\mu(\xi)\| = \|F_{\mu, \xi}\| \leq \|\xi\|$. We now verify the two properties of M_μ .

- (i) This equality follows directly from the defining property of the vector $M_{\mu, \xi}$.
- (ii) Using the equality $\mu(\chi_\Lambda) = 1$, for any $x \in E$ and $f \in E^*$ we have

$$\left\langle M_\mu\left(\bigoplus_\infty \{x : \lambda \in \Lambda\}\right), f \right\rangle = \mu\left(\bigoplus_\infty \{\langle x, f \rangle : \lambda \in \Lambda\}\right) = \langle x, f \rangle \mu(\chi_\Lambda) = \langle x, f \rangle.$$

Since $f \in E^*$ is arbitrary, we obtain the desired equality. $\square$

Proposition 3.5. *Let E be a reflexive Banach space. Let $(e_\lambda)_{\lambda \in \Lambda} \subset E$ and $(f_\lambda)_{\lambda \in \Lambda} \subset E^*$ be two families such that $\sup\{\|e_\lambda\| \|f_\lambda\| : \lambda \in \Lambda\}$ is finite. Then for any singular state μ on $\ell_\infty(\Lambda)$ and any bounded linear operator $\mathcal{T} : \mathcal{B}(E) \rightarrow \mathcal{B}(E)$ the map*

$$T_{\mu, \mathcal{T}} : E \rightarrow E, x \mapsto M_\mu\left(\bigoplus_\infty \{\mathcal{T}(x \circ f_\lambda)(e_\lambda) : \lambda \in \Lambda\}\right)$$

is a well-defined bounded linear operator with norm at most $\|\mathcal{T}\| \sup\{\|e_\lambda\| \|f_\lambda\| : \lambda \in \Lambda\}$.

Proof. For $x \in E$, denote $\xi_{\mathcal{T},x} = \bigoplus_{\infty} \{\mathcal{T}(x \circ f_{\lambda})(e_{\lambda}) : \lambda \in \Lambda\}$. Clearly,

$$\sup\{\|\xi_{\mathcal{T},x}(\lambda)\| : \lambda \in \Lambda\} \leq \|\mathcal{T}\|\|x\| \sup\{\|e_{\lambda}\|\|f_{\lambda}\| : \lambda \in \Lambda\},$$

so $\xi_{\mathcal{T},x} \in \bigoplus_{\infty} \{E : \lambda \in \Lambda\}$ and the value $M_{\mu}(\xi_{\mathcal{T},x})$ is well defined; hence so is the map $T_{\mu,\mathcal{T}}$. It is easy to check that $T_{\mu,\mathcal{T}}$ is linear. Furthermore, using proposition 3.4, for any $x \in E$ we get

$$\|T_{\mu,\mathcal{T}}(x)\| \leq \|M_{\mu}\|\|\xi_{\mathcal{T},x}\| \leq \|\mathcal{T}\|\|x\| \sup\{\|e_{\lambda}\|\|f_{\lambda}\| : \lambda \in \Lambda\}.$$

This proves the norm estimate for $T_{\mu,\mathcal{T}}$. $\square$

Proposition 3.6. *Let E be a reflexive Banach space with a Schauder basis $(e_{\lambda})_{\lambda \in \Lambda}$. Let $T : E \rightarrow E$ be a non-compact bounded linear operator. Then there exists a weakly null, semi-normalized block-basic sequence $(u_n)_{n \in \mathbb{N}}$ of $(e_{\lambda})_{\lambda \in \Lambda}$ such that $(T(u_n))_{n \in \mathbb{N}}$ is semi-normalized.*

Proof. Note that the property of a sequence to be weakly null, norm null or semi-normalized is preserved when passing to subsequences or equivalent sequences. We shall use these facts many times during the proof.

Since T is not compact, there exists a weakly null sequence $(x_n)_{n \in \mathbb{N}} \subset E$ such that $(T(x_n))_{n \in \mathbb{N}}$ is not norm null. Passing to a subsequence, we may assume that $\inf\{\|T(x_n)\| : n \in \mathbb{N}\} > 0$. Note that $(x_n)_{n \in \mathbb{N}}$ cannot be norm null, since otherwise $(T(x_n))_{n \in \mathbb{N}}$ would be norm null. Therefore, passing to a subsequence once again, we may additionally assume that $\inf\{\|x_n\| : n \in \mathbb{N}\} > 0$. As $(x_n)_{n \in \mathbb{N}}$ is weakly convergent, it is bounded; so is $(T(x_n))_{n \in \mathbb{N}}$, because T is bounded. Thus we have constructed a weakly null, semi-normalized sequence $(x_n)_{n \in \mathbb{N}}$ such that $(T(x_n))_{n \in \mathbb{N}}$ is also semi-normalized.

Since $(x_n)_{n \in \mathbb{N}}$ is a countable family, it lies in the closed linear span of some countable subfamily $(e_{\lambda})_{\lambda \in \Lambda'}$ of the basis $(e_{\lambda})_{\lambda \in \Lambda}$. By the Bessaga–Pełczyński selection principle [7], proposition 1.3.10] there exists a block-basic sequence $(u_k)_{k \in \mathbb{N}}$ of $(e_{\lambda})_{\lambda \in \Lambda'}$ which is equivalent to some subsequence of $(x_n)_{n \in \mathbb{N}}$. As $(u_k)_{k \in \mathbb{N}}$ is a block-basic sequence of $(e_{\lambda})_{\lambda \in \Lambda'}$, it is also a block-basic sequence of the larger basis $(e_{\lambda})_{\lambda \in \Lambda}$. Recall that $(x_n)_{n \in \mathbb{N}}$ is weakly null and semi-normalized, with $(T(x_n))_{n \in \mathbb{N}}$ also semi-normalized. The sequence $(u_k)_{k \in \mathbb{N}}$ inherits these properties, since it is equivalent to a subsequence of $(x_n)_{n \in \mathbb{N}}$. $\square$

Proposition 3.7. *Let Λ be a set, let $p \in (1, +\infty)$, and denote $E = \ell_p(\Lambda)$. Let $(e_{\lambda})_{\lambda \in \Lambda} \subset E$ be the standard basis and $(f_{\lambda})_{\lambda \in \Lambda} \subset E^*$ the respective biorthogonal functionals. Then for any singular state μ on $\ell_{\infty}(\Lambda)$ and any bounded linear operator $\mathcal{T} : \mathcal{B}(E) \rightarrow \mathcal{K}(E)$ the map $T_{\mu,\mathcal{T}}$ is a compact linear operator of norm at most $\|\mathcal{T}\|$.*

Proof. If E is finite-dimensional, then $\mathcal{B}(E) = \mathcal{K}(E)$, so there is nothing to prove.

Suppose E is infinite-dimensional, so that Λ is infinite, and denote $T := T_{\mu,\mathcal{T}}$. By proposition 3.5, the map T is a bounded linear operator with $\|T\| \leq \|\mathcal{T}\|$. Assume, towards a contradiction, that T is not compact. By proposition 3.6 we obtain a block-basic sequence $(u_n)_{n \in \mathbb{N}}$ of $(e_{\lambda})_{\lambda \in \Lambda}$ such that $c := \inf\{\|T(u_n)\| : n \in \mathbb{N}\} > 0$ and $C := \sup\{\|u_n\| : n \in \mathbb{N}\} < +\infty$.

Fix $n \in \mathbb{N}$. By the Hahn-Banach theorem there exists $g_n \in E^*$ such that $\|g_n\| = 1$ and $\langle T(u_n), g_n \rangle = \|T(u_n)\|$. Consider the vector $s_n := \bigoplus_{\infty} \{\langle \mathcal{T}(u_n \circ f_{\lambda})(e_{\lambda}), g_n \rangle : \lambda \in \Lambda\}$. Note that $s_n \in \ell_{\infty}(\Lambda)$, because $\sup\{|s_n(\lambda)| : \lambda \in \Lambda\} \leq C\|\mathcal{T}\|$. It is easy to check that $\mu(s_n) = \langle T(u_n), g_n \rangle$, so $|\mu(s_n)| = \mu(s_n) \geq c$. By proposition 3.3, the set $A_n := \{\lambda \in \Lambda : |s_n(\lambda)| > c/2\}$ is infinite. Since the sets A_n are infinite for all $n \in \mathbb{N}$, by induction we can construct a sequence of distinct elements $(\lambda_n)_{n \in \mathbb{N}} \subset \Lambda$ with $\lambda_n \in A_n$ for all $n \in \mathbb{N}$. In other words, there exists a sequence of distinct elements $(\lambda_n)_{n \in \mathbb{N}} \subset \Lambda$ such that $|\langle \mathcal{T}(u_n \circ f_{\lambda_n})(e_{\lambda_n}), g_n \rangle| > c/2$.

Since $(u_n)_{n \in \mathbb{N}}$ is a block basis of the standard basis of E , the supports of these vectors are disjoint, so for any $x \in \ell_\infty(\mathbb{N})$ and $y \in \ell_p(\Lambda)$ we get

$$\sum_{n \in \mathbb{N}} \|x(n) \langle y, f_{\lambda_n} \rangle u_n\| = \left(\sum_{n \in \mathbb{N}} |x(n)|^p |\langle y, f_{\lambda_n} \rangle|^p \|u_n\|^p \right)^{1/p} \leq C \|x\| \left(\sum_{n \in \mathbb{N}} |\langle y, f_{\lambda_n} \rangle|^p \right)^{1/p} \leq C \|x\| \|y\|.$$

This means that the series $\sum_{n \in \mathbb{N}} x(n) \langle y, f_{\lambda_n} \rangle u_n$ is absolutely convergent, and a fortiori convergent. Therefore, for each $x \in \ell_\infty(\mathbb{N})$ we obtain a well-defined map

$$S_x : E \rightarrow E, y \mapsto \sum_{n \in \mathbb{N}} x(n) (u_n \circ f_{\lambda_n})(y).$$

One can easily check that S_x is a linear operator, and the estimate above shows that $\|S_x\| \leq C \|x\|$. Now for each $n \in \mathbb{N}$ we define a map

$$f_n : \ell_\infty(\mathbb{N}) \rightarrow \mathbb{C}, x \mapsto \langle \mathcal{T}(S_x)(e_{\lambda_n}), g_n \rangle.$$

Clearly, it is a bounded linear functional. The sequence $(e_{\lambda_n})_{n \in \mathbb{N}}$ is weakly null, and by definition of $\mathcal{T}$ the operator $\mathcal{T}(S_x)$ is compact, so the sequence $(\mathcal{T}(S_x)(e_{\lambda_n}))_{n \in \mathbb{N}}$ is norm null. As the functionals $(g_n)_{n \in \mathbb{N}} \subset E^*$ are norm bounded, we conclude that

$$\lim_{n \rightarrow \infty} f_n(x) = \lim_{n \rightarrow \infty} \langle \mathcal{T}(S_x)(e_{\lambda_n}), g_n \rangle = 0.$$

This means that $(f_n)_{n \in \mathbb{N}}$ is a weak* null sequence. On the other hand,

$$\sum_{k \in \mathbb{N}} |f_n(\chi_{\{k\}})| \geq |f_n(\chi_{\{n\}})| = |\langle \mathcal{T}(u_n \circ f_{\lambda_n})(e_{\lambda_n}), g_n \rangle| > c/2.$$

So $\liminf_{n \rightarrow \infty} \sum_{k \in \mathbb{N}} |f_n(\chi_{\{k\}})| > 0$, which contradicts Phillips' lemma [[5], corollary 3.4]. Therefore T is compact. $\square$

Proposition 3.8. *Let Λ be a set, let $p \in (1, +\infty)$, and denote $E = \ell_p(\Lambda)$. Let $(e_\lambda)_{\lambda \in \Lambda} \subset E$ be the standard basis and $(f_\lambda)_{\lambda \in \Lambda} \subset E^*$ the respective biorthogonal functionals. Let μ be any singular state on $\ell_\infty(\Lambda)$. Define*

$$\tau_\mu : \mathcal{B}(\mathcal{B}(E), \mathcal{K}(E)) \rightarrow \mathcal{K}(E), \mathcal{T} \mapsto T_{\mu, \mathcal{T}}.$$

Then τ_μ is a contractive $\mathcal{B}(E)$ -morphism of right $\mathcal{B}(E)$ -modules such that $\tau_\mu \circ \rho_{\mathcal{K}(E)} = 1_{\mathcal{K}(E)}$, where $\rho_{\mathcal{K}(E)} : \mathcal{K}(E) \rightarrow \mathcal{B}(\mathcal{B}(E), \mathcal{K}(E)), K \mapsto (T \mapsto K \circ T)$.

Proof. By proposition 3.7, the map τ_μ is well defined. It is routine to check that τ_μ is linear. From the same proposition we also obtain that $\|\tau_\mu(\mathcal{T})\| \leq \|\mathcal{T}\|$, so τ_μ is contractive. For any $\mathcal{T} \in \mathcal{B}(\mathcal{B}(E), \mathcal{K}(E))$, $A \in \mathcal{B}(E)$ and $x \in E$ we have

$$\begin{aligned} \tau_\mu(\mathcal{T} \cdot A)(x) &= M_\mu \left(\bigoplus_{\infty} \{(\mathcal{T} \cdot A)(x \circ f_\lambda)(e_\lambda) : \lambda \in \Lambda\} \right) \\ &= M_\mu \left(\bigoplus_{\infty} \{\mathcal{T}(A(x) \circ f_\lambda)(e_\lambda) : \lambda \in \Lambda\} \right) \\ &= \tau_\mu(\mathcal{T})(A(x)) \\ &= (\tau_\mu(\mathcal{T}) \cdot A)(x). \end{aligned}$$

Therefore τ_μ is a right $\mathcal{B}(E)$ -module map. Finally, using proposition 3.4 and the biorthogonality of the families $(e_\lambda)_{\lambda \in \Lambda}$ and $(f_\lambda)_{\lambda \in \Lambda}$, we obtain, for all $K \in \mathcal{K}(E)$ and $x \in E$,

$$\begin{aligned}\tau_\mu(\rho_{\mathcal{K}(E)}(K))(x) &= M_\mu \left(\bigoplus_\infty \{ \rho_{\mathcal{K}(E)}(K)(x \circ f_\lambda)(e_\lambda) : \lambda \in \Lambda \} \right) \\ &= M_\mu \left(\bigoplus_\infty \{ \langle e_\lambda, f_\lambda \rangle K(x) : \lambda \in \Lambda \} \right) \\ &= M_\mu \left(\bigoplus_\infty \{ K(x) : \lambda \in \Lambda \} \right) \\ &= K(x).\end{aligned}$$

In other words, $\tau_\mu \circ \rho_{\mathcal{K}(E)} = 1_{\mathcal{K}(E)}$. □

Theorem 3.9. *Let E be an ℓ_p -space for some $p \in (1, +\infty)$. Then the right $\mathcal{B}(E)$ -module $\mathcal{K}(E)$ is 1-relatively injective.*

Proof. By assumption $E = \ell_p(\Lambda)$ for some set Λ . If Λ is finite, then E is finite-dimensional, so $\mathcal{K}(E) = \mathcal{B}(E)$, and this module is 1-relatively injective by Proposition 3.1. Assume now that Λ is infinite, and fix a singular state μ on $\ell_\infty(\Lambda)$, which exists by the discussion in Section 2. By proposition 3.8 there exists a contractive right $\mathcal{B}(E)$ -morphism τ_μ which is a left inverse for $\rho_{\mathcal{K}(E)} : \mathcal{K}(E) \rightarrow \mathcal{B}(\mathcal{B}(E), \mathcal{K}(E))$, $K \mapsto (T \mapsto K \circ T)$. By Helemskii's criterion and White's theorem (Section 2), the contractivity of τ_μ shows that the right $\mathcal{B}(E)$ -module $\mathcal{K}(E)$ is 1-relatively injective. □

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